

COAGULOPATIA X CIRROSE



Rodolpho Pedro

Clínica Médica e Terapia Intensiva – FMUSP

UTI da Gastroenterologia HCFMUSP + UTI do Hospital Vila Nova Star

Comitê de Gastrointensivismo – AMIB

Doutor em Ciências Médicas - HCFMUSP



Cirrótico na UTI

- 1 a cada 8 (12,5%) >> Sangramento relevante
- 75% dos sangramentos >> TGI
- > 90% dos sangramentos TGI >> Hipertensão portal



Perioperative thromboprophylaxis in liver transplant patients

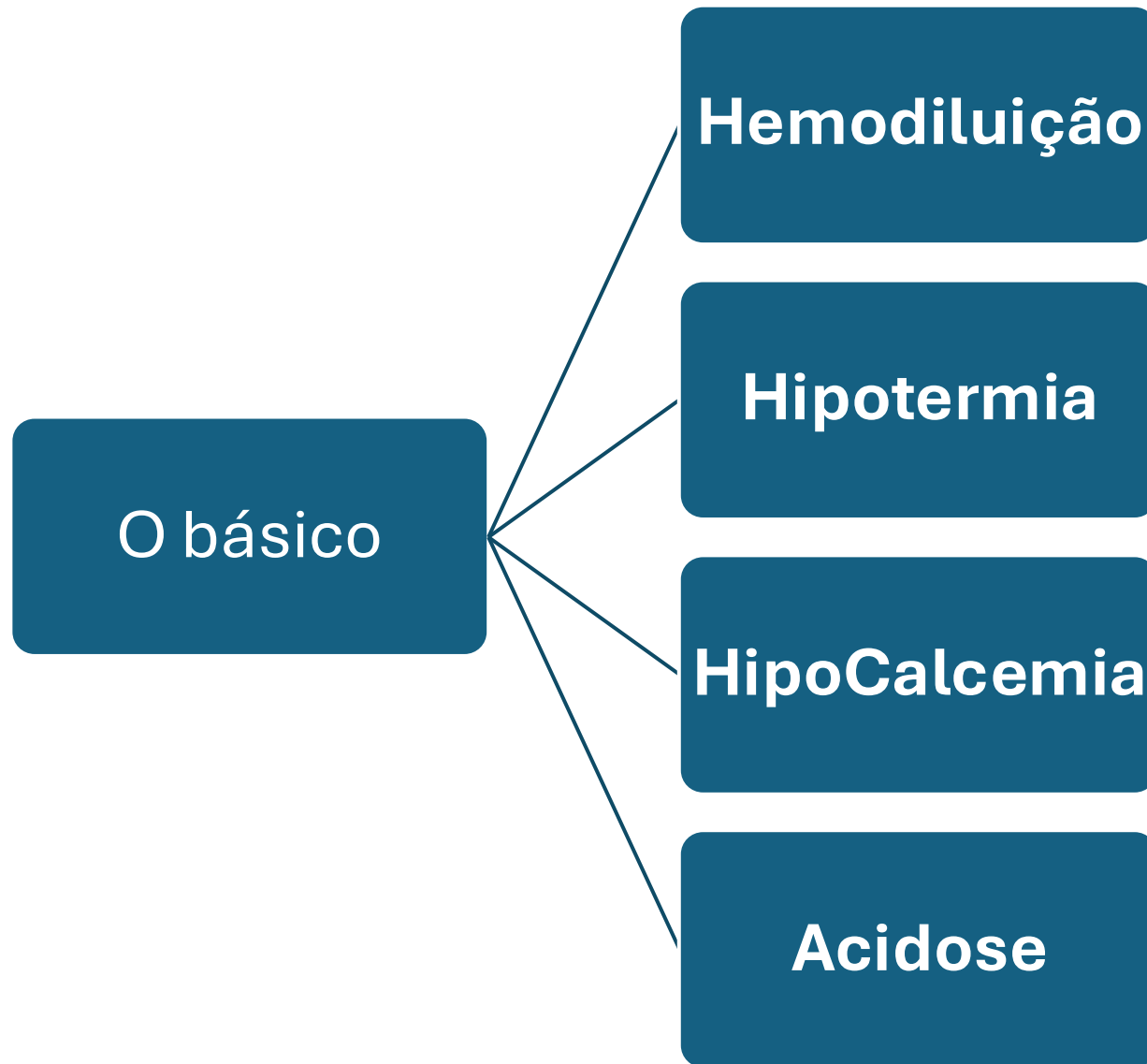
World J Gastroenterol 2018 July 21; 24(27): 2931-2948

NEW CONCEPT OF REBALANCED HEMOSTASIS AND THE UTILITY OF STANDARD LABORATORY TESTS

Table 1 Balance of antihemostatic and prohemostatic drivers in the cirrhotic patient

	Anti-hemostatic drivers	Pro-hemostatic drivers
Primary hemostasis	Abnormal platelet function Thrombocytopenia	Elevated vWF Reduced ADAMTS 13
Coagulation	Decreased production of thrombopoietin Reduced synthesis of factors II, V, VII, IX, X, and XI Vitamin K deficiency	Platelet hyperreactivity Elevated factor VIII Reduced anticoagulant protein C, protein S, antithrombin III
Fibrinolysis	Hypo-dysfibrinogenemia Low α 2-antiplasmin, factor XIII, and TAFI Elevated tPA	Procoagulant changes in fibrin structure Low plasminogen High PAI

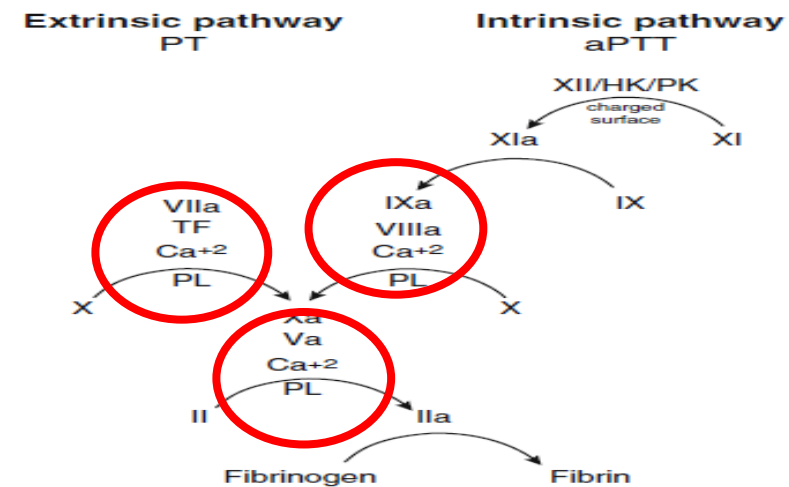




Coagulopatia em UTI

Hipocalcemia

Hemodiluição



Intraoperative Hypercoagulability During Liver Transplantation as Demonstrated by Thromboelastography

Dominik Krzanicki, Anita Sugavanam, and Susan Mallett

Department of Anaesthesia, Royal Free Hospital, London, United Kingdom

of standard plasma coagulation tests, as underlined in the Baveno VI guidelines^[21], comes from the fact they are performed without thrombomodulin addition, which is the main protein C activator, and they only detect the first 5% of whole thrombin formation. The



The coagulopathy of cirrhosis assessed by thromboelastometry and its correlation with conventional coagulation parameters

Armando Tripodi ^{a,*}, Massimo Primignani ^b, Veena Chantarangkul ^a, Ylenia Viscardi ^a, Alessandra Dell'Era ^{b,c}, Federica M. Fabris ^{b,c}, Pier Mannuccio Mannucci ^a

Table 2

Spearman correlation coefficients (p value) between thromboelastometry (CT, CFT or MCF) and parameters of coagulation.

Coagulation parameter	Thromboelastometry		
	CT	CFT	MCF
Platelets	-0.206 (0.16)	-0.741 (<0.001)	0.691 (<0.001)
Antithrombin	-0.022 (0.89)	-0.400 (0.005)	0.467 (0.001)
Fibrinogen	-0.023 (0.88)	-0.565 (<0.001)	0.590 (<0.001)
Factor II	-0.011 (0.94)	-0.343 (0.02)	0.392 (0.006)
PT-INR	0.264 (0.06)	0.356 (0.01)	-0.440 (0.001)

Bold types denote significant correlations.



ROTEM in the setting of liver transplant surgery reduces frozen plasma transfusion

Muntadhar Al Moosawi^a, Jacqueline Trudeau^{a,b}, Tyler Smith^{a,c}, Alexandre Lefebvre^d, Andrew W. Shih^{a,c,*}

^a Department of Pathology and Laboratory Medicine, University of British Columbia, Vancouver, British Columbia, Canada

^b Department of Anesthesiology, Pharmacology, and Therapeutics, University of British Columbia, Vancouver, British Columbia, Canada

^c Centre for Blood Research, University of British Columbia, Vancouver, British Columbia, Canada

^d Department of Cardiothoracic Anesthesiology, Copenhagen University Hospital, Copenhagen, Denmark

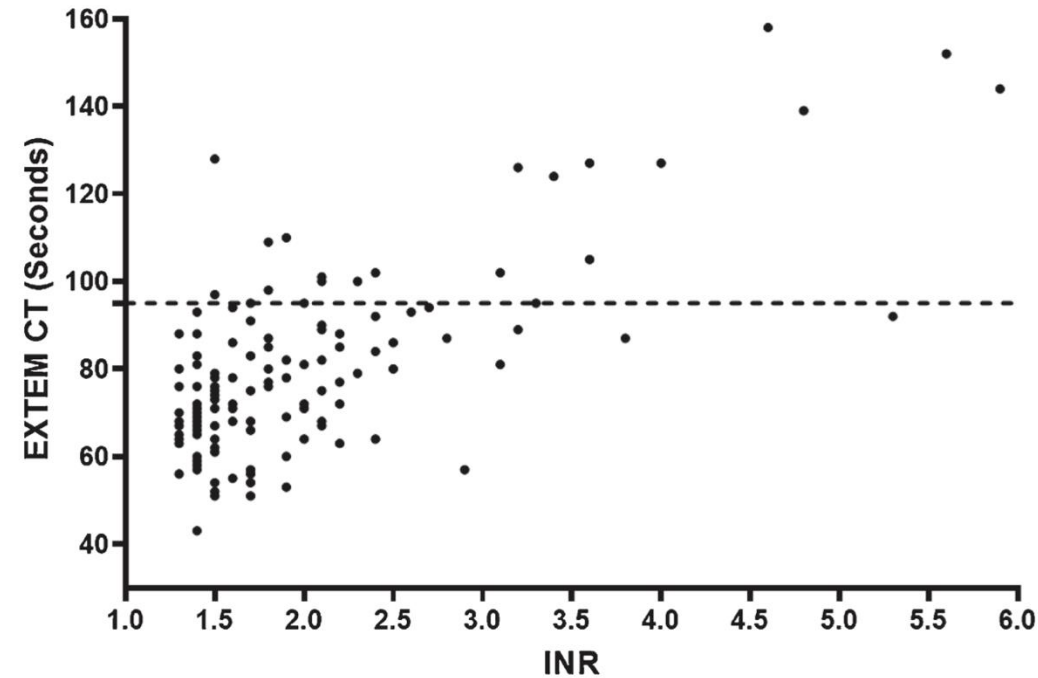


Fig. 1. Correlation between INR and EXTEM CT.

The horizontal line represents the upper limit of normal for the clotting time using the EXTEM reagent based on reference ranges used at our center.

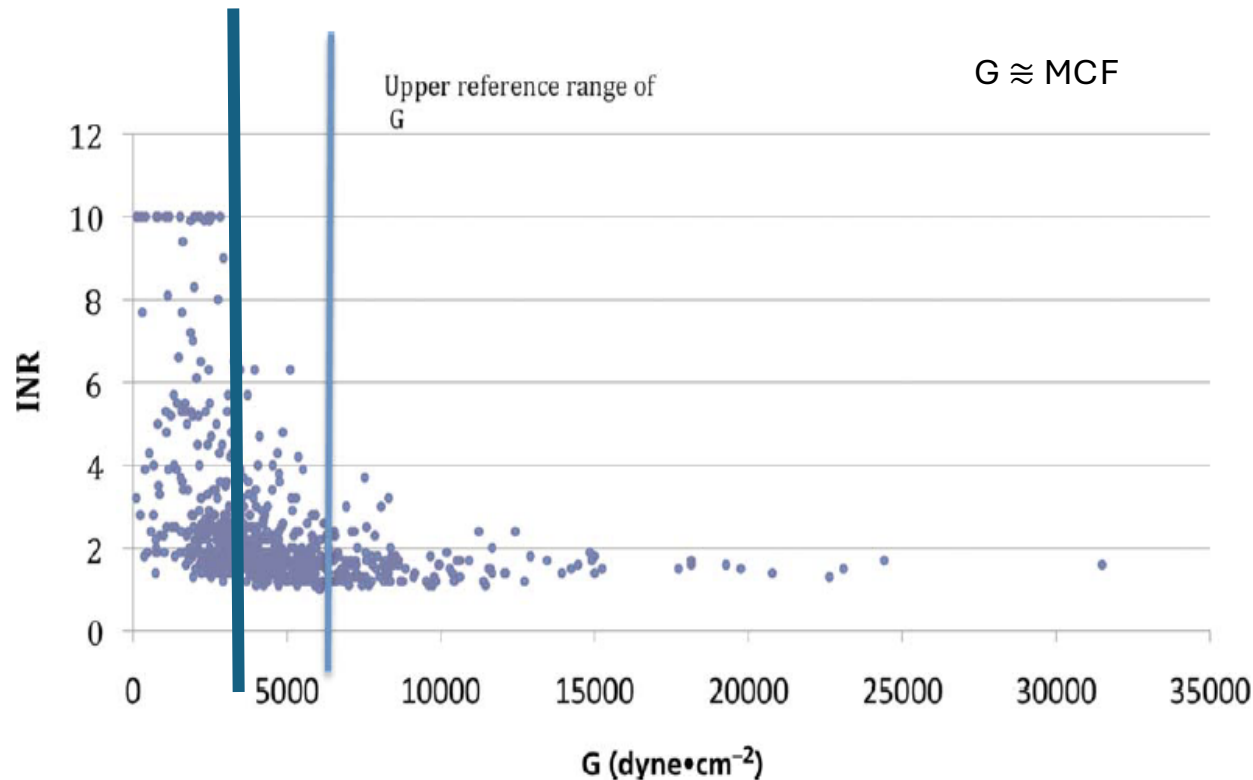
The correlation between INR and EXTEM CT improves only when INR is greater than 3.0 ($\rho = 0.703$).



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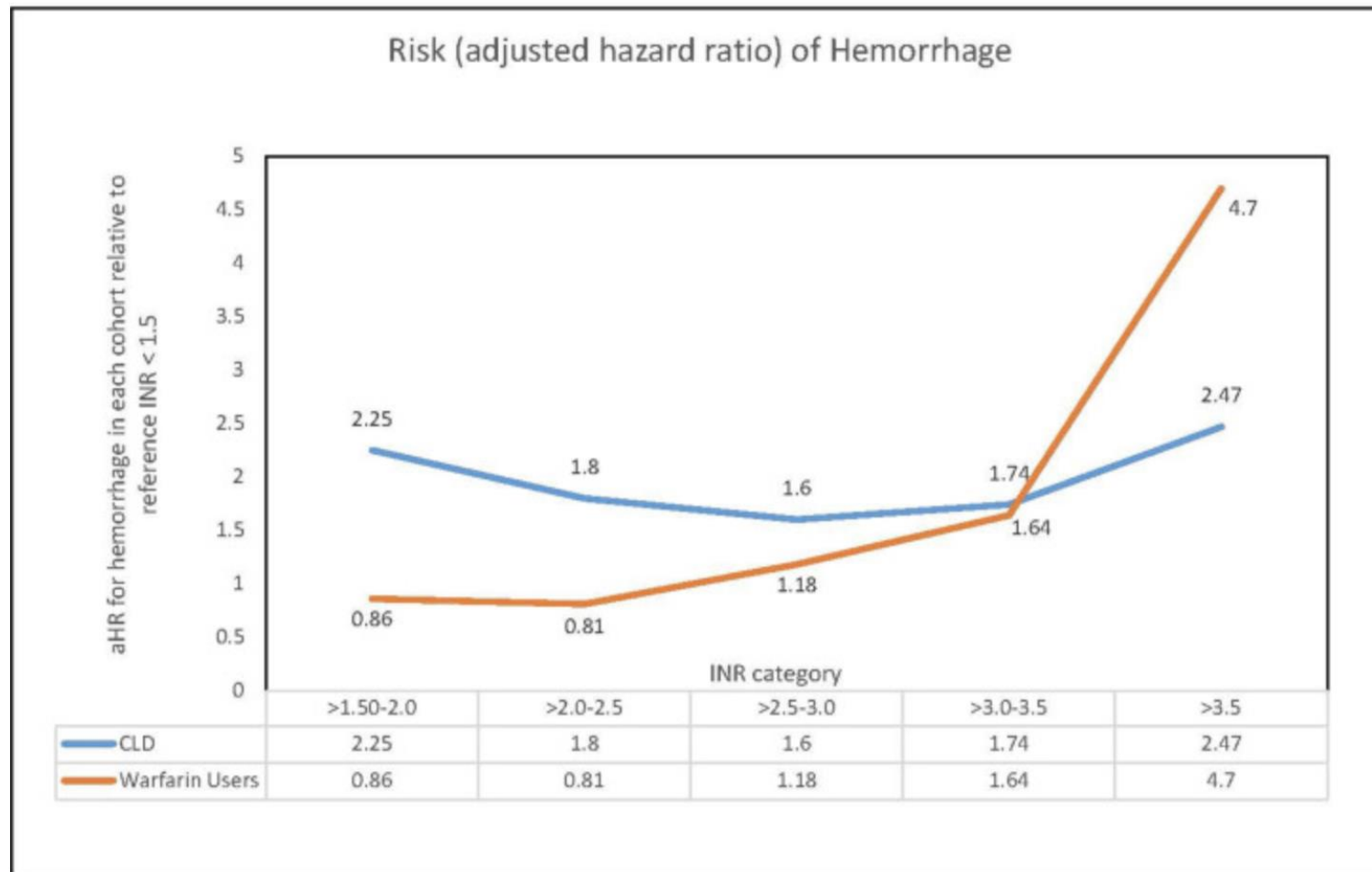
traces that were hypercoagulable (ie, above the normal reference range) could be associated with an INR between 0.9 and 3.8.



Different risks of hemorrhage in patients with elevated international normalized ratio from chronic liver disease versus warfarin therapy, a population-based retrospective cohort study

Amber Afzal   • Brian F. Gage • Luo Suhong • Martin W. Schoen • Kevin Korenblat • Kristen M. Sanfilippo

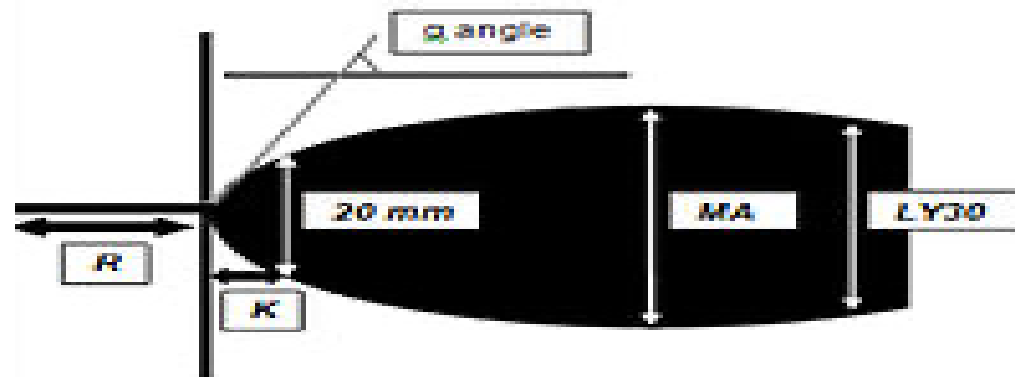
DOI: <https://doi.org/10.1111/jth.15743>



Minimal effects of acute liver injury/acute liver failure on hemostasis as assessed by thromboelastography

R. Todd Stravitz^{1,*}, Ton Lisman³, Velimir A. Luketic¹, Richard K. Sterling¹, Puneet Puri¹, Michael Fuchs¹, Ashraf Ibrahim², William M. Lee⁴, Arun J. Sanyal¹

Results: Thirty-seven of 51 patients (73%) had ALF and 22 patients (43%) underwent liver transplantation or died. Despite a mean INR of 3.4 ± 1.7 (range 1.5–9.6), mean TEG parameters were normal, and 5 individual TEG parameters were normal in 32 (63%). Low maximum amplitude, the measure of ultimate clot



<https://en.wikipedia.org/wiki/Thromboelastography>



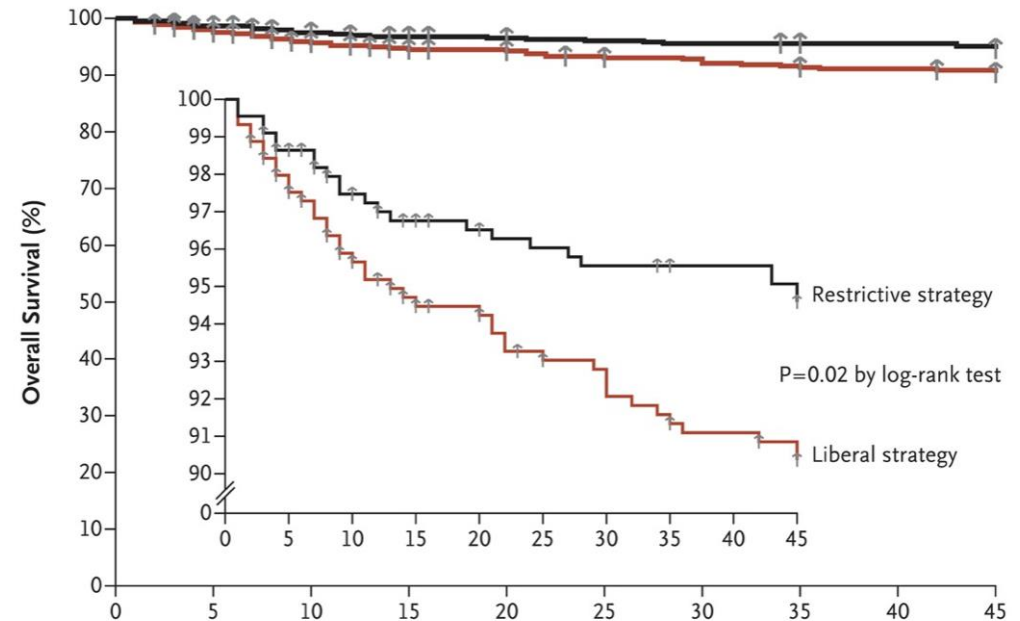
3.Suporte transfusional ?????

Mecânica x Coagulopatia

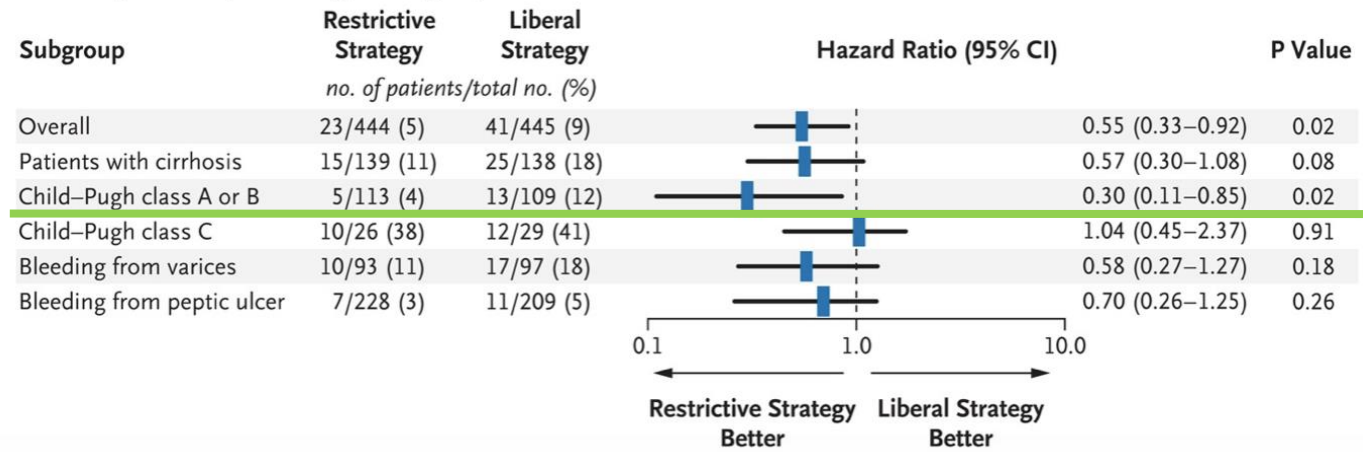
Alvos Hb 7-9 x > 9

- Menor Mortalidade
- Menor Sangramento
- Menor Pressão Portal

A Survival, According to Transfusion Strategy



B Death by 6 Weeks, According to Subgroup



0.45 to 2.37). Within the first 5 days, the portal-pressure gradient increased significantly in patients assigned to the liberal strategy (P=0.03) but not in those assigned to the restrictive strategy.



Liberal x Pressão Portal

Gut, 1974, 15, 99-101

The portal pressure-blood volume relationship in cirrhosis¹

DAVID S. ZIMMON AND RICHARD E. KESSLER

From the Medical and Surgical Services, Veterans Administration Hospital, and New York University School of Medicine, New York, New York

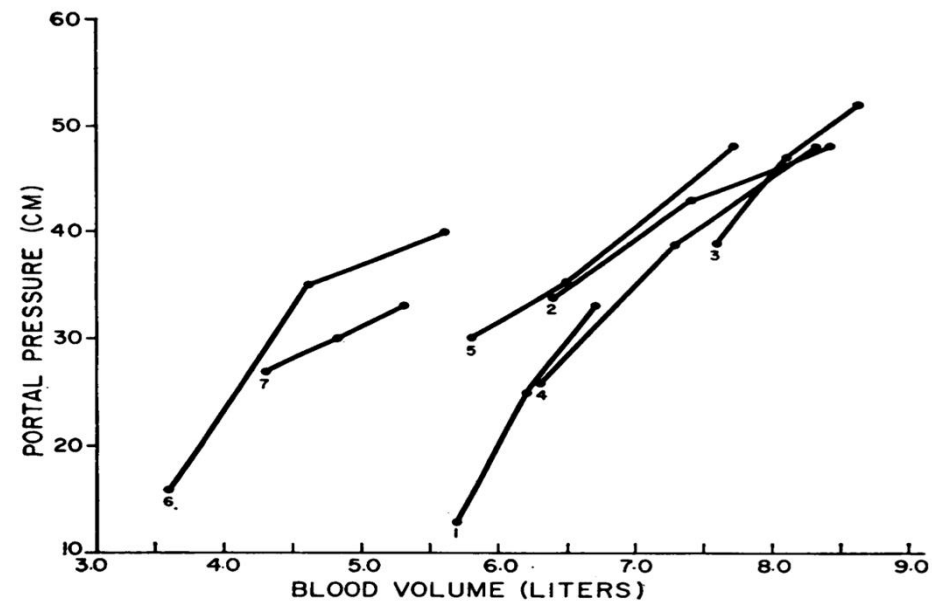


Fig. 1 *Portal pressure-blood volume curves from seven cirrhotic patients with portal hypertension derived from three measurements of portal pressure at blood volumes differing by more than 500 ml.*



Qual problema de tentar corrigir INR profilaticamente na fulminante?



Critério de Kings College	
Paracetamol	Não Paracetamol
pH arterial < 7,3 após ressuscitação volêmica	INR > 6.5
Ou todos esses: - INR > 6,5 - Cr > 3.4 - HE 3-4	Ou 3 de 5: - Drogas, Indeterminado ou Viral Não A, Não B - Icterícia – EH > 7 dias - Idade < 10 ou > 40 - INR > 3.5 - BT > 17.4



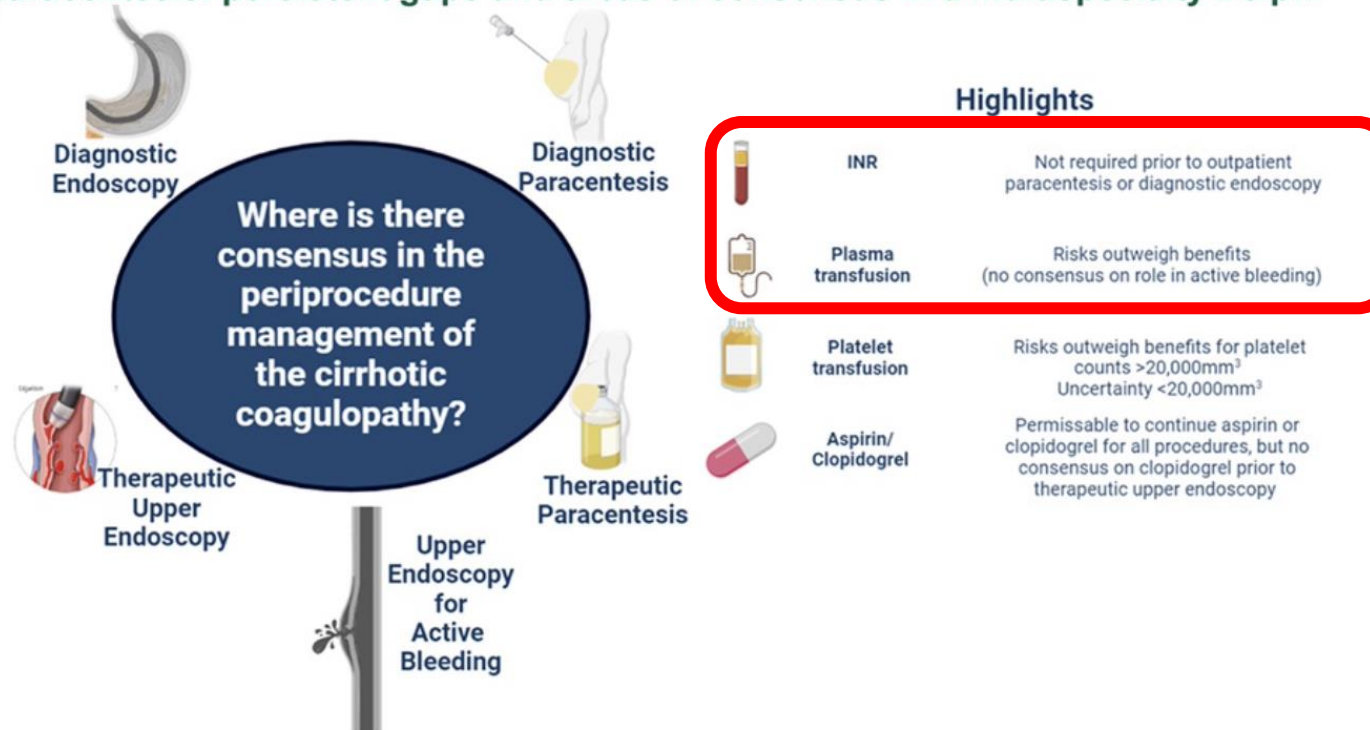
Management of coagulopathy among patients with cirrhosis undergoing upper endoscopy and paracentesis: Persistent gaps and areas of consensus in a multispecialty Delphi

Tapper, Elliot B.¹; Warner, Matthew A.²; Shah, Rajesh P.³; Emamaullee, Juliet⁴; Dunbar, Nancy M.⁵; Sholzberg, Michelle⁶; Poston, Jacqueline N.^{7,8}; Soto, Robin J.⁹; Sarwar, Ammar¹⁰; Pillai, Anjana¹¹; Reyner, Karina¹²; Mehta, Shivang¹³; Ghabril, Marwan¹⁴; Morgan, Timothy R.¹⁵; Caldwell, Stephen¹⁶

[Author Information](#)

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Management of coagulopathy among patients with cirrhosis undergoing upper endoscopy and paracentesis: persistent gaps and areas of consensus in a multispecialty Delphi



Sokol, et al. *Hepatology*.

HEPATOLOGY



Coagulopathy in cirrhosis: A prospective study to correlate conventional tests of coagulation and bleeding following invasive procedures in cirrhotics

Apurva Shah¹ • Deepak Amarapurkar¹ • Mrudul Dharod¹ • Madhuri Chandnani¹ •
Rajiv Baijal² • Praveen Kumar² • Mayank Jain³ • Nikhil Patel⁴ • Praful Kamani⁵ •
Sonali Gautam¹ • Nimish Shah² • Sandeep Kulkarni² • Soham Doshi²

Table 2 Invasive procedures and clinically significant bleeding during invasive procedures

Variable	Low risk procedures [#] , <i>n</i> (%)	High risk procedures [*] , <i>n</i> (%)
Clinically significant bleeding		
Group A	0	3 (17)
Group B	0	0
<i>p</i> -value	-\$	0.061 ^{\$\$}

[#] Low risk procedures include abdominal paracentesis, endoscopic band ligation, endoscopic glue injection and endoscopic sclerotherapy for gastroesophageal varices

^{*} High risk procedures include central venous cannulation, liver biopsy, polypectomy, transjugular intrahepatic portosystemic shunt, bone marrow

^{\$\$} *p*-value by Fisher’s exact test

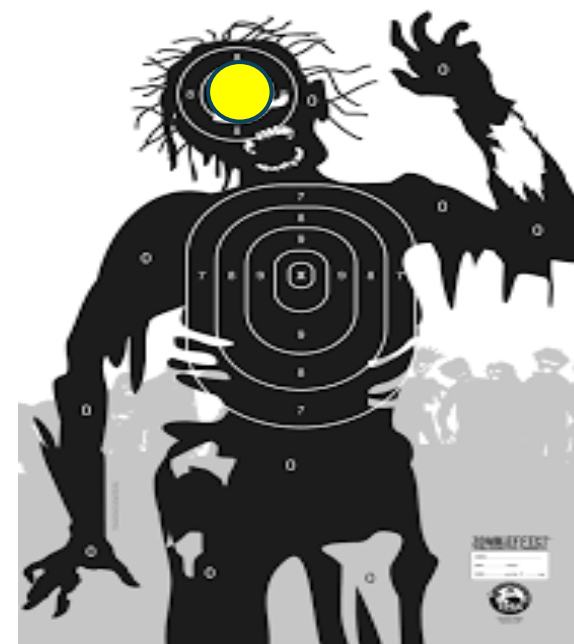
Grupo A: INR > 1,5

Group A
(*n*=128)

Group B
(*n*=252)

Transfusão Maciça X Guiada por TEG

- Custo-Benefício?
- Disponibilidade



Thromboelastography-Guided Blood Component Use in Patients With Cirrhosis With Nonvariceal Bleeding: A Randomized Controlled Trial

Manoj Kumar ✉, Juned Ahmad, Rakhi Maiwall, Ashok Choudhury, Meenu Bajpai, Lalita G. Mitra, Vandana Saluja, Prashant Mohan Agarwal, Chhagan Bihari ... See all authors ✓

First published: 31 May 2019 | <https://doi.org/10.1002/hep.30794> | Citations: 62

TABLE 3. Distribution of Blood Products Transfused and Transfusion-Related Side Effects in TEG and SOC Groups During 42 Days of Follow-Up

Variable	TEG Group (n = 49)	SOC Group (n = 47)	P Value
Total amount of FFP infused (mL)	20,860	40,300	<0.001
FFP (mL) infused/patient	440 (0-1,320)	880 (0-1,640)	<0.001
Total amount of platelet pools infused (U)	26	71	<0.001
Platelet pools (U) infused/patient	1 (0-1)	2 (0-3)	<0.001
Total amount of cryoprecipitate infused (U)	278	814	<0.001
Cryoprecipitate (U) infused/patient	4 (0-24)	16 (4-36)	<0.001

Transfused RBCs	40 (81.6)	35 (74.5)	0.464
Total amount of RBCs transfused (packs)	118	126	0.584
RBCs transfused (packs) per patient	2 (0-7)	2 (0-10)	0.584
Any serious transfusion-related reaction	15 (30.6)	35 (74.5)	<0.001
TRALI	6 (12.2)	23 (48.9)	<0.001
TACO	5 (10.2)	10 (21.3)	0.166
ARDS	1 (2)	9 (17)	0.011
Other serious transfusion reactions	3 (6.1)	13 (27.7)	0.006





Thromboelastography-Guided Transfusion Decreases Intraoperative Blood Transfusion During Orthotopic Liver Transplantation: Randomized Clinical Trial

S.-C. Wang, J.-F. Shieh, K.-Y. Chang, Y.-C. Chu, C.-S. Liu, C.-C. Loong, K.-H. Chan, S. Mandell, and M.-Y. Tsou

Conclusion. Thromboelastography-guided transfusion decreases transfusion of fresh-frozen plasma in patients undergoing orthotopic liver transplantation, but does not affect 3-year survival.



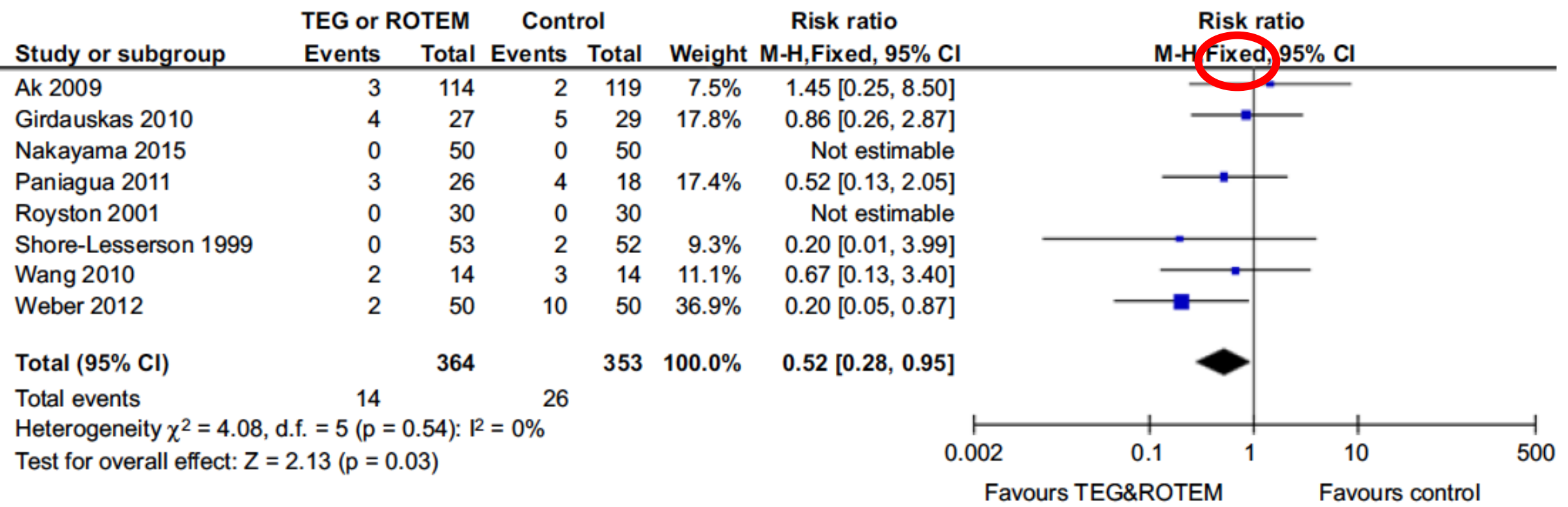


Figure 3 Forest plot of primary outcome: mortality at longest follow-up.

being zero-event trials; when we changed from a fixed model to random effects model for the meta-analysis, the statistical significance was no longer present. Addi-



Rotational thromboelastometry-guided blood component use in cirrhotic children undergoing invasive procedures: Randomized controlled trial

Liver International. 2022;42:2492–2500.

Arjun Maria¹ | Bikrant B. Lal¹ | Rajeev Khanna¹ | Vikrant Sood¹ |
Amar Mukund² | Meenu Bajpai³ | Seema Alam¹

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WILEY | 2497

TABLE 2 Comparison of the blood components transfusion requirement in the two groups

Parameter	ROTEM group (N = 30)	Conventional group (N = 30)	Odds ratio (95% CI)	Significance
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TABLE 3 Comparison of procedure-related bleed, transfusion-related complications and cost incurred in the two groups

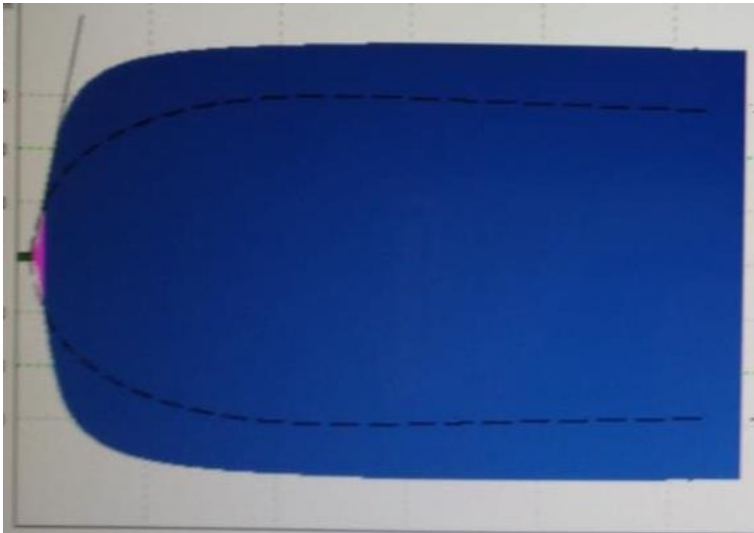
Parameter	ROTEM-based transfusion group (N = 30)	Conventional tests-based transfusion group (N = 30)	p value
Procedure-related bleeds	1 (3.3%)	2 (6.7%)	1



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Department of Anaesthesia, Royal Free Hospital, London, United Kingdom

**Trombo
Hipercoagulante**



Among the 117 primary transplants (ie, with the exclusion of the 7 regraft procedures), there were 6 cases (5%) of HAT. Four of the 6 required a regraft procedure as a result of HAT. Three of the 6 cases had high G traces during their initial transplants [$P=0.25$ (χ^2)], and 4 of the 6 cases demonstrated shortened R times [$P=0.79$ (χ^2)].

20% na Pop. Geral x 66% nos casos de HAT

Como avaliar o ROTEM ???

Próxima Aula !

